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Pinning AeST's Free Background Parameter: $\mathcal{Q}_0 \approx 3 \times 10^{-3} - 1.5 \times 10^{-2}$ Mpc^{-1} from Galaxy-Scale Phenomenology

Carl P. Zimmerman, Briar Creek Tech ORCID 0009-0008-3508-7982 2026-08-13 (v4, 2026-08-14: the drain reading settled — the soft low edge becomes operative; the recombination liability priced at order-of-magnitude grade; and a fixed-point consistency result — the pin survives its own strongest internal challenge, at the cost of a candidate transport mechanism. All three additions adversarially refereed before filing.)

Abstract

The Aether-Scalar-Tensor theory (AeST; Skordis & Złośnik 2021, PRL **127** 161302) reproduces the Planck CMB power spectrum and the galaxy-scale radial acceleration relation, but leaves the scalar background rate $\mathcal{Q}_0 \equiv d\varphi/dt$ free — its authors state explicitly that the dark-sector density is "not (classically) predicted." We show that a framework built on AeST with three additional commitments — the dark-sector charge carrying the full dark-matter abundance ($\rho = \mathcal{Q}_0$ n), the selection $\beta = 1$, and w = -1 exactly — collapses that freedom to a **one-parameter family**, and that galaxy-scale phenomenology fixes the remaining coordinate. The result is

> $\mathcal{Q}_0 \approx 2.4 \times 10^{-3} - 1.5 \times 10^{-2}$ **Mpc^{-1}**, equivalently $\mathbf{K''(\mathcal{Q}_0) \approx 10^2 - 10^3}$, *(v4: the low edge is 2.4×10^{-3} under the now-settled steady-state-continuity reading of the local charge density — Section 5's "soft by 1.4×" bullet became the operative edge; the v1–v3 low edge 3.4×10^{-3} was the residence reading)*,

a region strictly interior to AeST's published fits: it lies 45–215× above their exponential fit and 5–20× below their cosh fit, and *excludes* the Higgs-type fit which Skordis and Złośnik themselves describe as "incompatible with a MOND limit." Expressed in the natural variable $\mu \equiv \sqrt{(K''(\mathcal{Q}_0))}$, **both of AeST's MOND-compatible published parameter sets fall inside the pinned band (μ = 122 and 195, band 33–1295) while the MOND-incompatible one does not (μ = 41231)** — an independent corroboration, since no CMB information entered the derivation.

We are explicit about what this is not. The pin is an order-of-magnitude *class* statement, not an equality: the two profiles it equates have different radial shapes and the implied $\mathcal{Q}_0$ drifts by a factor 2.1–2.7 across 3.7–60 kpc. It does not price the theory's nonlinear sector at recombination,

and cannot: AeST's authors note that a_0 "does not appear in the linear cosmological regime but will play a role once nonlinear terms from $\mathcal{F}(\mathcal{Y},\mathcal{Q})$ kick in," which is precisely the regime at issue. Every quantitative claim is reproduced by committed scripts.

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1. What AeST leaves free

AeST is the relativistic MOND-class theory that fits the CMB. Its action is

$$S = \int d^4x\sqrt{-g}\left[\frac{R-2\Lambda_{\rm bare}}{16\pi G} + \mathcal{L}_{\rm aether}[A^\mu, g] + \mathcal{F}(Y,Q)\right] + S_{\rm matter},$$

with the invariants $Q \equiv A^\mu\nabla_\mu\varphi$ and $Y \equiv (g^{\mu\nu} + A^\mu A^\nu)\nabla_\mu\varphi\nabla_\nu\varphi$ built from a unit-timelike aether A^μ and a scalar φ. The physical content sits in the free function $\mathcal{F}$. Skordis and Złośnik exhibit three choices (cosh, Higgs-type, exponential) that fit Planck, with fitted parameters spanning **four orders of magnitude** in the background rate $\mathcal{Q}_0$: 10^{-4}, 10^{-1} and 1 Mpc^{-1} respectively. Nothing in the theory selects among them.

2. The three commitments that collapse the freedom

The framework studied here adds to AeST:

1. $\rho = \mathcal{Q}_0$ **n identically**, with the shift charge carrying the *full* dark-matter abundance. This is not an assumption of convenience — it is forced by the shift-symmetric condensate structure, and it is the same property that makes w = -1 exact.
2. $\beta = 1$, the selection fixing the dark sector's scale (the same selection that yields the derived evolution law a_0(z)).
3. **w = -1 exactly**, up to $\mathcal{O}(\nu_0^2)$ with $\nu_0 \lesssim 1.8 \times 10^{-4}$.

Together with the framework's acceleration-scale normalisation a_0 = κc √(Gρ_Λ) = c² √(Λ/32π) = 9.3619×10^{-11} m/s², these give

$$\mathcal{Q}_0 = \frac{\sqrt{\Lambda}\, R_{\rm dm}}{\nu_0\, \mu}, \qquad R_{\rm dm} \equiv \Omega_{\rm dm}/\Omega_\Lambda = 0.387, \qquad \mu \equiv \sqrt{K''(\mathcal{Q}_0)}.$$

This is one equation in two unknowns. The abundance chain therefore does not fix $\mathcal{Q}_0$; it reduces AeST's freedom to a one-parameter family, parameterised by the dimensionless combination

$$X \equiv \frac{\mathcal{Q}_0 c^2}{a_0}, \qquad \text{equivalently} \qquad \mathcal{Q}_0\,[\text{Mpc}^{-1}] = \frac{X}{31112}.$$

3. Fixing the remaining coordinate

The framework's kinematic identity $u_\mu = -\nabla_\mu\varphi/s$ — aether-independent, and used nonlinearly elsewhere in the corpus — slaves the condensate flow speed to the scalar gradient. The same gradient must simultaneously supply the static MOND force at radial-acceleration-relation radii. One field, one gradient, two jobs: an equality, which fixes X.

Using AeST's own quasi-static diagonalisation ($\Phi = \Phi + \varphi$ with Φ Newtonian), the gate variable is $y = [(g_{\rm tot} - g_N)/a_0]^2$, and evaluating it against the framework's own kernel across 3.7–60 kpc with the committed accretion free-fall speed gives

	X	$\mathcal{Q}_0$ [Mpc^{-1}]	$\mu = \sqrt{}(\mathrm{K}'')$
a_0-line kernel	106 – 323	0.0034 – 0.0104	33 – 570
MS08 operative kernel	120 – 453	0.0039 – 0.0146	40 – 800
defensible envelope	70 – 1340	0.0022 – 0.0431	33 – 1295

4. The corroboration

Nothing above used CMB information. Converting AeST's three *published* fits into the same variable $\mu = \sqrt{}(\mathrm{K}''(\mathcal{Q}_0))$ (noting that their exponential function carries $\mathrm{K}'' = 4\mathrm{K}_2$ rather than $2\mathrm{K}_2$):

AeST fit	K_2	$\mu = \sqrt{}(\mathrm{K}''(\mathcal{Q}_0))$	inside the pinned band?	MOND-compatible?
cosh	7.5×10^3	**122**	**yes**	yes
exponential	9.5×10^3	**195**	**yes**	yes
Higgs-type	8.5×10^8	41231	no	**no** — SZ21: “incompatible with a MOND limit”

Both MOND-compatible published fits land inside the band derived from galaxy-scale phenomenology alone; the fit their own authors exclude on MOND grounds lands outside it. We regard this as supporting evidence of the pin's *class*, not as a measurement.

5. What this does not establish

Stated as plainly as the result:

- **It is a class, not an equality.** The static-MOND and drain profiles have different radial shapes, so exact simultaneity is impossible at any single X; the implied $\mathcal{Q}_0$ drifts by 2.1–2.7× across 3.7–60 kpc. This is a real residual, not a rounding.
- **The pin inherits a CANDIDATE grade.** It rests on one adversarially-refereed lane. The identification of the corpus's $\mathcal{Q}_0$ with AeST's (units, cosmic time coordinate, background and kinetic normalisations) is derivation-grade and was verified on five legs against the published source; the *value* is not yet at that grade.
- **The nonlinear sector at recombination is now priced at order-of-magnitude grade (v4), and the price is small.** At the pinned $\mathcal{Q}_0$ the theory's $\mathcal{Y}$-sector is genuinely active at $z \approx 1090$ (flow-y $\approx$ 0.5–280 across the band — the premise is confirmed, not evaded). But the lever arm on the dust mode is the vector–scalar mixing share, which scales with $\mathcal{Q}_0$ and is 0.05–0.3% at the pinned values (0.5% at the defensible-envelope corner); the kinetic-coefficient deviation is bounded by 1, and the $\mathcal{Y}$-branch's direct stress is $\sim 5 \times 10^{-14}$

of the dust density. Product: $\leq$ **0.2% on the core band,** $\leq$ **0.5% at the envelope** — sub-percent everywhere this note defends. The full nonlinear Boltzmann computation remains owed for derivation grade, but it is now a confirmation run, not an existential one (`nbody_2026/stage62_cmb_horn_oom_2026.py`).

- $\mathbf{a_0}$ **in this framework is a *local* quantity** — $\mathcal{A}(\mathrm{Q}) = \mathrm{a}_0^2 = \kappa^2\mathrm{G}(\text{-K}(\mathcal{Q}))$ depends on the local charge density, and the pin's radii (3.7–60 kpc) sit *inside* halos, where that density is orders above the cosmic mean. The pin is nonetheless $\mathbf{a_0}$**-free algebraically**: combining X = $\sqrt{}$y·c/v with y = [(g_tot-g_N)/a_0]^2 and $\mathcal{Q}_0$ = X·a_0/c^2 gives

$$\mathcal{Q}_0 = \frac{g_{\text{tot}} - g_N}{c\,v},$$

in which a_0 cancels identically — the pin is (MOND excess acceleration)/(c × drain speed), a statement in observables. The residual a_0-dependence enters only through the kernel that builds g_tot from g_N, is sub-linear (≈ S^0.5 in deep MOND, S^0.8 near y = 1), and is partly opposed by v_ff ∝ (GMa_0)^$\frac{1}{4}$ carrying the same factor in the same direction.

- **The low-edge softness is resolved (v4): the steady-state-continuity reading is operative.** For the framework's own committed picture — smooth capture, cold radial infall, no halt — mass conservation makes ρ = /(4πr^{2}v) exact; the residence-time estimate was that same argument with an inconsistent (virialised) shape. The 0.0024 Mpc^{-1} edge is therefore the operative one and is carried in the abstract band (`nbody_2026/stage61_drain_reading_fork_2026.py`). The defensible envelope of Section 3 is unchanged.

- **A fixed-point consistency result (v4), one each way.** The corpus's candidate "transport channel" (an outward shift-charge flux that could have drained the halo's captured charge) turns out, at this note's pinned $\mathcal{Q}_0$, to carry a flux that would gate the very accretion flow the pin's equality is built on: a strong drain would un-calibrate its own coefficient. **Favourable:** the pin survives its strongest internal challenge — the only self-consistent closure is the one that leaves the pin's flow intact. **Adverse:** that same argument demotes the transport channel from viable-candidate to conditional-dead, closing (at closure grade) a door the framework had open on its galaxy-scale dark-matter problem (`nbody_2026/stage63_cell3_transport_1p1d_2026.py`).

- **Consistency note, in the framework's favour:** the same effect is what fixes the framework's own ν_0 ceiling — the requirement that the local a_0^2 shift stay $\lesssim$ 1% through the drain is precisely the RAR bound on a local a_0, so this is a priced effect rather than an unpriced one.

- $\kappa = \frac{1}{2}$ **remains measured, not derived** (0.551 ± 0.043 by the distance-free method), and the coefficient's origin is unresolved. Nothing here changes that.

- $\mathbf{K_2}$ **is pinned only to ~3 decades**, because the ν_0 window carries a factor 8.3 that propagates quadratically. λ_s remains free.

6. Why it is worth recording

AeST's parameter freedom in $\mathcal{Q}_0$ is normally regarded as irreducible — its authors say so. The claim here is narrow and checkable: *if* one commits to the dark-sector charge carrying the full dark-matter abundance with w = -1 exact, then galaxy-scale phenomenology alone selects a $\mathcal{Q}_0$ region interior to the published CMB fits, and specifically the region occupied by the MOND-compatible ones. That is a statement about AeST's parameter space, testable by anyone with the theory and a galaxy sample, and falsifiable in the obvious way: a CMB fit demanding $\mathcal{Q}_0$ outside 10^{-3}–10^{-1} Mpc^{-1} would break it.

Reproducibility

All claims are reproduced by committed scripts at https://github.com/carlzimmerman/zimmerman-formula:

- `nbody_2026/stage58_x_to_q0_2026.py` — the conversion identity, the pinned band, the corroboration table, and the consequences (9 checks).
- `nbody_2026/stage58_ev_q0pin_lane_2026.py` — the adversarial lane's own trial script (23 checks).
- `nbody_2026/stage56_xpin_verdict_2026.py` — the X-pin and the closure of its named escape route.
- `nbody_2026/stage57_sz21_corrected_refile_2026.py` — the AeST parameter records and the background identity SRC = 3f_dust H/$\mathcal{Q}_0$.
- `nbody_2026/stage59_local_a0_verdict_2026.py` — the local-a_0 adjudication behind $§§$5's first three bullets (7 checks), including the symbolic cancellation.
- `nbody_2026/stage61_drain_reading_fork_2026.py` — the drain-reading settlement behind the v4 operative low edge (10 checks).
- `nbody_2026/stage62_cmb_horn_oom_2026.py` — the order-of-magnitude pricing of the recombination liability (7 checks).
- `nbody_2026/stage63_cell3_transport_1p1d_2026.py` — the fixed-point consistency result (8 checks, including the referee-corrected supply-gating mechanism).
- `RETRACTIONS.md` — the standing scope-and-retraction record, including the withdrawal of an earlier version of the recombination-side analysis (an error in the framework's *disfavour*, corrected).

Prior art and attribution. AeST is Skordis & Złośnik, PRL **127** 161302 (arXiv:2007.00082); the interpolation kernel used here is Milgrom & Sanders 2008, ApJ **678** 131, Eq. (13) at $\alpha = \frac{1}{2}$. The framework's contribution is the acceleration-scale normalisation and the three commitments of Section 2, not the underlying theory.